$$\begin{aligned}
\langle \dot{\sigma}_x \rangle &= -\omega \langle \sigma_y \rangle \\
\langle \dot{\sigma}_y \rangle &= \omega \langle \sigma_x \rangle \\
\langle \dot{\sigma}_z \rangle &= 0. \qquad (4.27)
\end{aligned}$$

Does this look familiar? If not, go back to *Volume I*, Lecture 10. There, we studied the classical rotor in a magnetic field. The equations were exactly the same, except that instead of expectation values, we were studying the actual motion of a deterministic system. Both there and here, the solution is that the 3-vector-operator $\vec{\sigma}$(or the 3-vector $\vec{L}$ in *Volume I*) precesses like a gyroscope around the direction of the magnetic field. The precession is uniform, with angular velocity ω.

This similarity to classical mechanics is very pleasing, but it's important to take note of the difference. Exactly *what* is precessing? In classical mechanics, it's just the *x* and *y* components of angular momentum. In quantum mechanics, it's an expectation value. The expectation value for a σ_z measurement does not change with time, but the other two expectation values do. Regardless, the result of each individual measurement of each spin component is still either +1 or −1.

Exercise 4.4: Verify the commutation relations of Eqs. 4.26.

4.12 SOLVING THE SCHRÖDINGER EQUATION

The iconic Schrödinger equation that appears on T-shirts has this form:

$$i\hbar \frac{\partial \Psi(x)}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x)}{\partial x^2} + U(x)\Psi(x).$$